

The K-Means clustering algorithm is applied to group similar agricultural patterns based on environmental features. The Elbow Method is used to determine the optimal number of clusters by analyzing WCSS values. The model is trained using the `fit()` and `fit_predict()` methods to identify meaningful crop groupings for intelligent agricultural analysis.

Elbow Graph: The Elbow Graph is used to determine the optimal number of clusters by plotting the Within-Cluster Sum of Squares (WCSS) against different cluster values and identifying the point where the rate of decrease sharply changes, known as the “elbow point.” It is demonstrated below:

Training the model: The selected machine learning algorithm is trained using the prepared training dataset to learn patterns and make accurate predictions based on input features.